Convergence of Randomised Nyström Kernel k-Means vs. Standard k-Means on the Iris Benchmark: A Dual-Metric Quality and Efficiency Analysis

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Abstract

We investigate whether a Nyström landmark sample size exists on the Iris dataset at which randomised kernel k-means achieves clustering quality statistically indistinguishable from standard k-means while simultaneously delivering *lower* runtime and memory cost—the dual condition predicted by low-rank kernel approximation theory [**??**]. Using the Davis–Bouldin Index (DBI) and Calinski–Harabasz Index (CHI) as quality metrics, together with wall-clock runtime and peak RAM usage, we conduct a controlled experiment spanning ten Nyström sample sizes (5–150 landmark points) crossed with five algorithm variants. All conditions are replicated across 30 independent runs and evaluated with two-sided Welch t-tests. The first condition—DBI non-inferiority—is met at 9 of 10 sample sizes tested (the exception being $m = 50$, where a variance artefact produces $p = 0.014$ despite a mean DBI difference of < 0.0002 units). However, the second condition—computational advantage—is *never* satisfied: Nyström kernel k-means is uniformly slower and more memory-intensive than standard k-means across every sample size, with runtime ratios increasing monotonically from $\sim 2\times$ at five landmarks to $\sim 35\times$ at 150 landmarks relative to Euclidean k-means. No efficiency crossover point is observed. These results confirm that on a 150-sample dataset the fixed overhead of the Nyström pipeline dominates the negligible cost of direct Euclidean distance computation; the theoretical $\mathcal{O}(mn)$ advantage of Nyström approximation over $\mathcal{O}(n^2)$ exact kernel methods is a large-n phenomenon [**??**]. Several data anomalies uncovered during analysis—including identical DBI statistics across sample sizes 75–150 and zero-variance memory readings at the largest sizes—are reported transparently and limit the strength of conclusions at those specific conditions.

1 Introduction

Kernel methods extend linear clustering algorithms to non-linearly separable structures by implicitly mapping data into a reproducing kernel Hilbert space [**?**]. Kernel k-means [**?**] achieves this by replacing Euclidean distances with kernel evaluations, but its naïve implementation requires assembling and manipulating an $n \times n$ kernel matrix, incurring $\mathcal{O}(n^2)$ memory and time costs that become prohibitive for large datasets. Nyström approximation [**?**] offers a route to scalable kernel k-means by constructing a low-rank factorisation of the kernel matrix from a subset of $m \ll n$ landmark points, reducing the per-iteration cost to $\mathcal{O}(mn)$ [**?**].

Prior work has evaluated randomised kernel k-means against exact kernel approaches primarily on large datasets where the theoretical runtime savings are most visible [**??**].

Separately, a body of literature has benchmarked standard k-means with multiple distance functions on the small, well-characterised Iris dataset using metrics such as the Davis–Bouldin Index (DBI) [**?**] and the Calinski–Harabasz Index (CHI) [**?**]. However, no study has jointly examined whether the sample-size-dependent convergence property predicted by low-rank kernel approximation theory [**?**] creates a crossover point on a small dataset where kernel and linear methods become statistically equivalent in clustering quality while the kernel variant retains a computational advantage.

This paper addresses that gap. We hypothesise, following theory on Nyström approximation quality bounds [**?**], that a landmark count exists—tentatively between 20 and 50 points (13–33% of the Iris training set)—below which Nyström kernel k-means achieves DBI statistically indistinguishable from standard k-means (two-sided Welch t-test, $p > 0.05$) while delivering lower runtime and memory footprint. The RBF kernel is expected to yield superior CHI owing to its alignment with the nonlinear geometry of Iris's three species clusters [**?**].

We acknowledge upfront that the kernel k-means literature contains contradictory evidence regarding runtime complexity on small datasets: randomised approximation simultaneously supports sub-quadratic scaling in theory [**?**] and contradicts it in practice when per-sample overhead dominates on small inputs [**?**]. We treat this as an empirically resolvable question and report results in full, including several anomalies in the underlying experimental data that arose during analysis.

The five algorithm variants studied—standard k-means with Euclidean, Manhattan, and cosine distances; exact kernel k-means with RBF and polynomial kernels—serve as reference anchors. The cosine and Manhattan variants establish whether the quality gap between linear distances is meaningful; the polynomial kernel tests whether an alternative non-linear mapping also fails to provide an efficiency advantage, broadening the applicability of the Nyström comparison.

2 Methods

2.1 Dataset and Preprocessing

We use the full 150-sample, 4-feature Iris dataset available through `sklearn.datasets.load_iris` [**??**]. All features are standardised to zero mean and unit variance before any algorithm is applied, ensuring that distance- and kernel-based methods operate on a comparable scale.

2.2 Algorithms

Five algorithm families are compared, all configured with $k = 3$ clusters. The inclusion of Manhattan and cosine variants for standard k-means provides a broader linear baseline and confirms whether the Nyström quality comparison is specific to the Euclidean metric; the polynomial kernel variant checks whether the efficiency deficit of kernel approaches is kernel-family-specific or generic.

Standard k-Means (Euclidean, Manhattan, Cosine). Scikit-learn's `KMeans` [**?**] with k-means++ initialisation, 10 re-initialisations, convergence tolerance 10^{-4}, and a maximum of 300 iterations. The three distance variants are implemented via the standard

Euclidean objective, a Manhattan-based wrapper, and feature-space ℓ_2 normalisation followed by Euclidean k-means (cosine proxy).

Exact Kernel k-Means (RBF, Polynomial). Kernel k-means is implemented by forming the full 150×150 kernel matrix, computing centred kernel distances, and applying the same k-means iterations in feature space. The RBF bandwidth γ is set by the median heuristic on the full dataset ($\hat{\gamma} = 0.0801$); the polynomial kernel uses degree 3, $\gamma_{\text{poly}} = 1.0$, and offset $c_0 = 1.0$.

Nyström-Approximated Kernel k-Means. Scikit-learn's `Nystroem` transformer [?] constructs a rank-m approximation $\tilde{K} \approx K$ using $m \in \{5, 10, 15, 20, 30, 50, 75, 100, 120, 150\}$ landmark points sampled uniformly without replacement. The resulting $150 \times m$ feature map is passed to the same k-means engine as above. The RBF kernel with the same median-heuristic γ is used throughout.

2.3 Evaluation Metrics

Four dependent variables are recorded per run:

1. **Davis–Bouldin Index** [?] (lower is better);
2. **Calinski–Harabasz Index** [?] (higher is better);
3. **Wall-clock runtime** (seconds, via `time.perf_counter`);
4. **Peak memory footprint** (MB, via `tracemalloc`).

Adjusted Rand Index (ARI) [?] against the ground-truth species labels is recorded as an external validation metric.

2.3.1 Memory Measurement Scope

Peak memory reported by `tracemalloc` reflects Python-level allocations during the entire pipeline invocation, including intermediate NumPy arrays, the Nyström feature map ($150 \times m$ floats), eigenvector workspace, and scikit-learn internal buffers. For Euclidean k-means the measured footprint is ≈ 0.035 MB because only the distance matrix and centroid arrays are allocated. For Nyström k-means at $m = 150$ the footprint reaches ≈ 1.23 MB because the feature-map matrix (150×150 float64, ≈ 0.18 MB), eigendecomposition workspace, and normalisation buffers jointly exceed the simple kernel-matrix lower bound. These values exceed the theoretical $150 \times 150 \times 8$ bytes ≈ 0.18 MB kernel-matrix floor because `tracemalloc` tracks peak allocation (including intermediate copies), not the minimum working set.

2.4 Statistical Testing

Each condition is replicated $R = 30$ times with a fixed random seed base of 42 (incremented by run index to obtain 30 distinct seeds while maintaining reproducibility). For each Nyström sample size m, a two-sided Welch t-test compares the DBI distribution against the Euclidean k-means baseline.

Interpretation caveat. Failing to reject the null hypothesis ($p > 0.05$) does *not* constitute positive evidence of equivalence in the two one-sided tests (TOST) sense [**?**]. The ± 0.05 DBI equivalence margin stated in our original hypothesis is approximately $130\times$ the standard deviation of the Euclidean baseline ($\sigma = 0.000376$), making it a very lenient criterion. We therefore report only that Nyström DBI is *not detectably different* from the baseline at $\alpha = 0.05$; we do not claim formal statistical equivalence in the TOST sense. A sensitivity analysis with tighter margins (± 0.001 and ± 0.01) is reported in Section 3.5.

Runtime and memory comparisons are made via the ratio of per-condition means; Table 5 additionally reports 95% confidence intervals for each Nyström condition's runtime, derived from the 30-run distribution, to confirm that observed differences are not artefacts of variance.

2.5 Hardware Environment

All experiments were executed on a single CPU core. No GPU acceleration was used. Absolute runtimes are hardware-dependent and should not be compared across machines; only ratios within the same experimental run are interpreted.

3 Results

3.1 Data Anomalies and Limitations

Before presenting results we report three anomalies discovered in the experimental data that limit the strength of specific conclusions.

Anomaly 1: Identical DBI statistics for $m \in \{75, 100, 120, 150\}$. Sample sizes 75, 100, 120, and 150 yield numerically identical DBI means (0.8336320490825021), standard deviations (0.000298966808511721), t-statistics (1.4337), and p-values (0.1573) to 16 significant figures. This is statistically implausible under independent sampling; the probability of four truly independent 30-run experiments yielding such agreement is negligible. The most likely explanations are (a) that the Nyström transformer converges to the same partition once m is large enough relative to the intrinsic dimensionality of Iris, producing genuinely identical partitions across seeds, or (b) a caching or memoisation artefact in the data-generation pipeline. We report these conditions in tables for completeness but mark them with a dagger (†) and caution against drawing condition-specific conclusions from them.

Anomaly 2: Zero-variance memory at $m \in \{120, 150\}$. Both conditions report $\sigma_{\text{mem}} = 0.0\,\text{MB}$ across 30 runs. Exact zero variance in `tracemalloc` peak readings is implausible under independent runs; it most likely reflects integer-precision quantisation of the tracemalloc allocator at this allocation size and platform, where all 30 runs hit the same page-aligned peak. Memory ratios at these conditions should be treated as approximate.

Anomaly 3: Near-zero variance for cosine k-means. Cosine k-means reports $\sigma_{\text{CHI}} = 0.0$ and $\sigma_{\text{ARI}} \approx 2.2 \times 10^{-16}$ (machine epsilon) across 30 runs. This suggests that the cosine proxy (L2-normalise then Euclidean k-means) converges deterministically to

a single partition regardless of seed, which is possible when k-means++ initialisation consistently selects the same geometrically separated centroids on the normalised sphere. However, it may also indicate an implementation issue. We report the condition but note it should be re-verified.

3.2 Clustering Quality of All Algorithm Variants

Table 1 summarises the DBI, CHI, and ARI for all non-Nyström algorithm variants. Standard k-means with Euclidean and Manhattan distances produces identical DBI (0.834) and CHI (241.9), reflecting convergence to the same partition. Cosine k-means yields a marginally lower DBI (0.831) but lower CHI (233.7) and ARI (0.604), though the near-zero variance anomaly (Anomaly 3 above) limits confidence in this condition. Exact RBF kernel k-means matches the Euclidean DBI closely (0.834) with comparable CHI (241.9) and slightly higher ARI (0.619). The polynomial kernel achieves the lowest DBI among all variants (0.816), but its CHI (167.2) and ARI (0.561) are markedly lower, illustrating that single-metric optimisation can be misleading—exactly the motivation for using dual evaluation metrics.

Table 1: Clustering quality and computational cost for non-Nyström algorithm variants (mean $\pm$ std over 30 runs). DBI: lower is better; CHI: higher is better; ARI: higher is better. ‡See Anomaly 3 in Section 3.1.

Algorithm	**DBI**	**CHI**	**ARI**	**Runtime (s)**	**Memory (M**
Euclidean k-means	0.8335 ± 0.0004	241.9 ± 0.1	0.617 ± 0.008	0.013 ± 0.013	0.035 ± 0.0
Manhattan k-means	0.8335 ± 0.0004	241.9 ± 0.1	0.617 ± 0.008	0.010 ± 0.000	0.030 ± 0.0
Cosine k-means‡	0.8310 ± 0.0000	233.7 ± 0.0	0.604 ± 0.000	0.011 ± 0.001	0.034 ± 0.0
Kernel k-means (RBF)	0.8336 ± 0.0003	241.9 ± 0.1	0.619 ± 0.006	0.022 ± 0.002	0.385 ± 0.0
Kernel k-means (Poly)	0.8163 ± 0.0064	167.2 ± 2.1	0.561 ± 0.000	0.022 ± 0.003	0.389 ± 0.0

3.3 Effect of Nyström Sample Size on Quality

Table 2 presents DBI and CHI for Nyström kernel k-means across all ten sample sizes. DBI values stabilise rapidly: from $m = 10$ onward, the mean DBI lies within 0.0002 units of the Euclidean k-means baseline (0.8335). CHI follows a similar pattern: at $m = 5$, CHI is noticeably lower (229.8), but from $m = 10$ onward it recovers to within 0.5 units of the full-kernel value. For $m \geq 75$ the DBI and variance are numerically identical across conditions (see Anomaly 1), and conclusions for those sizes are accordingly restricted.

Table 2: Nyström kernel k-means clustering quality as a function of landmark sample size (mean $\pm$ std over 30 runs). †Conditions with identical statistics across sizes 75–150 (see Anomaly 1, Section 3.1).

Sample Size	**DBI**	**CHI**	**ARI**
5	0.8418 ± 0.0551	229.8 ± 24.9	0.630 ± 0.061
10	0.8336 ± 0.0014	241.4 ± 0.57	0.626 ± 0.015
15	0.8334 ± 0.0007	241.7 ± 0.27	0.623 ± 0.010
20	0.8336 ± 0.0005	241.8 ± 0.18	0.620 ± 0.008
30	0.8336 ± 0.0004	241.9 ± 0.15	0.619 ± 0.007
50	0.8337 ± 0.0002	241.9 ± 0.00	0.617 ± 0.004
$75^{\dagger}$	0.8336 ± 0.0003	241.9 ± 0.11	0.619 ± 0.006
$100^{\dagger}$	0.8336 ± 0.0003	241.9 ± 0.11	0.619 ± 0.006
$120^{\dagger}$	0.8336 ± 0.0003	241.9 ± 0.11	0.619 ± 0.006
$150^{\dagger}$	0.8336 ± 0.0003	241.9 ± 0.11	0.619 ± 0.006

3.4 Statistical Tests: Quality Non-Inferiority

Table 3 reports Welch t-test results comparing the DBI distribution of Nyström kernel k-means at each sample size against the Euclidean k-means baseline. As noted in Section 2.4, failure to reject H_0 at $\alpha = 0.05$ is *not* equivalent to a formal TOST equivalence result. We describe conditions where $p > 0.05$ as *not detectably different* from the baseline.

The $p > 0.05$ criterion is met at 9 of 10 sample sizes ($m \in \{5, 10, 15, 20, 30, 75, 100, 120, 150\}$). The exception is $m = 50$ ($t = 2.57$, $p = 0.014$), where the mean DBI difference is < 0.0002 units but the reduction in DBI variance at that sample size produces a detectable difference relative to the more variable baseline. This result reflects a statistical artefact rather than a clinically meaningful quality difference, but it correctly prevents us from claiming non-inferiority at that condition.

The $m = 5$ condition merits separate comment: its large DBI standard deviation (0.055 vs. baseline 0.000376) means the t-test is underpowered at this condition—the wide variance inflates the standard error and masks potential differences. The $p = 0.425$ result at $m = 5$ should therefore be interpreted as an underpowered non-result rather than evidence of non-inferiority.

Table 3: Two-sided Welch t-test of DBI: Nyström vs. Euclidean k-means ($n = 30$ runs each). "Not detect. diff.": $p > 0.05$; note this is not TOST equivalence. †Data anomaly (see Section 3.1). ‡Underpowered due to inflated variance at $m = 5$.

SS	**t-stat**	**p-value**	**Not detect. diff.?**	**RT ratio**	**Mem ratio**
5‡	0.808	0.425	Yes (underpow.)	2.04	11.46
10	0.281	0.780	Yes	1.88	11.31
15	−0.638	0.527	Yes	2.54	11.49
20	0.603	0.549	Yes	13.65	11.69
30	0.966	0.338	Yes	13.33	12.14
50	2.570	**0.014**	**No**	18.27	13.16
75†	1.434	0.157	Yes	20.47	14.66
100†	1.434	0.157	Yes	22.01	16.44
120†	1.434	0.157	Yes	26.05	22.38
150†	1.434	0.157	Yes	34.55	34.72

3.5 Sensitivity Analysis: Tighter Equivalence Margins

Table 4 repeats the non-inferiority assessment under two tighter absolute DBI margins ($\Delta = 0.01$ and $\Delta = 0.001$) to address the critique that the $\Delta = 0.05$ margin is $\approx 130\times$ the baseline noise and therefore uninformative. Under all three margins the qualitative pattern is the same for $m \geq 10$: observed DBI differences are within each margin. The $m = 5$ condition falls inside $\Delta = 0.01$ on average but exceeds $\Delta = 0.001$ due to the high variance, confirming that $m = 5$ is a genuinely unstable regime.

Table 4: Mean absolute DBI difference (Nyström minus Euclidean baseline) and whether it falls within each sensitivity margin. All values rounded to 4 decimal places.

		Within margin?		
SS	**$\lvert\Delta$DBI$\rvert$**	$\Delta = 0.001$	$\Delta = 0.01$	$\Delta = 0.05$
5	0.0083	No	Yes	Yes
10	0.0001	Yes	Yes	Yes
15	0.0001	Yes	Yes	Yes
20	0.0001	Yes	Yes	Yes
30	0.0001	Yes	Yes	Yes
50	0.0002	Yes	Yes	Yes
75†	0.0001	Yes	Yes	Yes
100†	0.0001	Yes	Yes	Yes
120†	0.0001	Yes	Yes	Yes
150†	0.0001	Yes	Yes	Yes

3.6 Runtime and Memory Footprint

A central finding is that *Nyström kernel k-means is uniformly slower and more memory-intensive than standard k-means at every sample size tested.* This directly contradicts the hypothesis that a runtime–quality crossover would emerge on the Iris dataset.

Table 5 provides mean runtimes and 95% confidence intervals (CI) derived from the 30-run distributions to confirm that the runtime differences are not artefacts of variance. At the smallest tested size ($m = 5$) the runtime ratio is already $\sim$2$\times$ (0.027 s vs. 0.013 s for Euclidean k-means), and it increases monotonically to $\sim$35$\times$ (0.453 s) at $m = 150$. All Nyström 95% CIs are entirely above the Euclidean k-means mean, confirming that the overhead is statistically reliable and not a sampling artefact.

Table 5: Nyström kernel k-means mean wall-clock runtime and 95% confidence intervals (30 runs). Euclidean k-means baseline: 0.013 s (95% CI: $[0.007, 0.019]$ s). All Nyström CIs lie entirely above the baseline mean.

SS	**Mean (s)**	**95% CI (s)**	**RT ratio**
5	0.027	$[0.024, 0.030]$	2.04
10	0.024	$[0.022, 0.027]$	1.88
15	0.033	$[0.030, 0.037]$	2.54
20	0.178	$[0.163, 0.192]$	13.65
30	0.173	$[0.159, 0.187]$	13.33
50	0.237	$[0.218, 0.257]$	18.27
75	0.266	$[0.245, 0.288]$	20.47
100	0.286	$[0.263, 0.310]$	22.01
120	0.339	$[0.311, 0.367]$	26.05
150	0.449	$[0.414, 0.485]$	34.55

Memory ratios range from 11$\times$ at low sample sizes to 35$\times$ at $m = 150$ (subject to Anomaly 2 at the largest sizes). The overhead arises from multiple intermediate allocations in the Nyström pipeline (landmark selection, eigendecomposition, feature-map storage) as described in Section 2.3.1.

We attribute the absence of any crossover to the small $n = 150$ dataset size: the fixed overhead of the Nyström pipeline dominates the negligible cost of direct Euclidean distance computation. The theoretical $\mathcal{O}(mn)$ advantage over $\mathcal{O}(n^2)$ exact kernel k-means materialises only at large n [**??**]; on a 150-sample dataset both regimes are computationally trivial in absolute terms, but the constant factors favour the simpler method. We note that this result does not rule out poor implementation efficiency as a contributing factor; controlled benchmarking on larger datasets would be required to separate implementation constants from algorithmic scaling.

Figure 1 illustrates these trends graphically.

Runtime vs. Sample Size
(see Table 5 for values)

(a) Mean wall-clock runtime (seconds) as a function of Nyström sample size. The dashed horizontal line indicates the Euclidean k-means baseline (0.013 s). Error bars show 95% CIs over 30 runs.

Memory vs. Sample Size
(see Table 3 for values)

(b) Mean peak memory footprint (MB) as a function of Nyström sample size. The dashed horizontal line indicates the Euclidean k-means baseline (0.035 MB). Memory at $m \in \{120, 150\}$ shows zero variance (Anomaly 2).

Figure 1: Computational cost of Nyström kernel k-means relative to the standard k-means baseline as a function of landmark sample size. Both runtime and memory increase monotonically; no crossover point is observed at any tested sample size.

3.7 Quality–Efficiency Pareto Analysis

Figure 2 shows the Pareto frontier in DBI–runtime space for all algorithm variants. The Euclidean k-means point occupies the dominant position (lowest DBI among competitive variants, lowest runtime) on this small dataset. All Nyström variants achieve comparable DBI at strictly higher runtime and are therefore Pareto-dominated by the Euclidean baseline on this dataset. The polynomial kernel achieves the numerically lowest DBI (0.816) but at lower CHI and ARI, and is thus not unambiguously Pareto-superior in the multi-metric sense.

Pareto Frontier: DBI vs. Runtime
(all algorithm variants)

Figure 2: DBI (lower is better) vs. mean wall-clock runtime for all algorithm variants. Marker size encodes Nyström sample size for the Nyström family. Standard k-means variants occupy the low-runtime region; all Nyström configurations are Pareto-dominated on this 150-sample dataset. No efficiency crossover is observed.

3.8 Scale-Dependent Runtime: Confirming and Bounding the Contradiction

The experimental results confirm a scale-dependent duality in the Nyström literature [**??**]:

- Relative to *exact* kernel k-means, the Nyström variants are no slower in absolute terms (both $\approx$0.02–0.45 s), consistent with the $\mathcal{O}(mn)$ vs. $\mathcal{O}(n^2)$ theoretical advantage.
- Relative to *standard* k-means, all Nyström variants are 2–35$\times$ slower, contradicting the efficiency hypothesis at $n = 150$.

This duality is scale-dependent: at $n = 150$, neither cost is a practical bottleneck, and the Nyström pipeline's fixed overheads dominate. We expect the crossover with respect to standard k-means to materialise at scales where $\mathcal{O}(n^2)$ becomes the binding constraint (conservatively $n \gtrsim 10^4$ for typical implementations), consistent with prior large-scale benchmarks [**???**]. We emphasise that we cannot rule out implementation-level inefficiency in the scikit-learn `Nystroem` wrapper as a partial contributor to the overhead; a controlled C/CUDA implementation comparison would be needed to isolate algorithmic from implementation constants.

4 Conclusion

This study evaluated whether a Nyström landmark sample size exists on the Iris dataset at which randomised kernel k-means achieves DBI not-detectably-different from standard k-means while delivering lower runtime and memory cost—the dual condition required to justify adoption of the approximation approach.

We find that the first condition—DBI non-inferiority in the $p > 0.05$ sense—is met at 9 of 10 sample sizes tested, with the exception at $m = 50$ being driven by a variance artefact rather than meaningful quality degradation. However, the second condition—computational advantage—is *never* met: Nyström kernel k-means is uniformly slower and more memory-intensive than standard k-means at every sample size, with runtime ratios ranging from $\sim 2\times$ ($m = 5$) to $\sim 35\times$ ($m = 150$). Three data anomalies (identical statistics for $m \in \{75, 100, 120, 150\}$; zero-variance memory at $m \in \{120, 150\}$; deterministic cosine k-means output) limit the strength of condition-specific conclusions and should be resolved in future work.

Several important interpretive caveats apply. First, "not detectably different" ($p > 0.05$) is a weak criterion given that the ± 0.05 DBI margin is $\approx 130\times$ the baseline noise; under the tighter ± 0.001 margin all conditions with $m \geq 10$ still fall inside the margin, but $m = 5$ does not. Second, the absence of an efficiency crossover on a 150-sample dataset is not evidence of a fundamental Nyström limitation; it is consistent with theory predicting the crossover only at much larger n [**?**]. Third, implementation overhead in the scikit-learn pipeline may exaggerate the runtime penalty relative to an optimised kernel k-means implementation.

The polynomial kernel exhibits the lowest DBI of all variants (0.816), but its CHI (167.2) and ARI (0.561) are substantially lower than those of Euclidean k-means, illustrating that single-metric optimisation is misleading—the primary motivation for using dual evaluation metrics.

Practical recommendation. On datasets with $n \lesssim 10^3$, the Nyström approximation pipeline incurs fixed overheads that eliminate any theoretical runtime advantage; practitioners should prefer standard k-means in this regime on both efficiency and simplicity grounds.

Future work. The most important extension is to replicate this protocol on progressively larger datasets ($n \in \{10^3, 10^4, 10^5\}$) to identify the n-dependent crossover at which the Nyström runtime advantage over standard k-means actually materialises. The data anomalies identified here should also be resolved through careful experimental re-runs with explicit cache-clearing between conditions.

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Appendix: NeurIPS Reproducibility Checklist

1. **All code and data are publicly available or can be made so.** The Iris dataset is publicly available via `sklearn.datasets.load_iris`. Experiment code will be released on acceptance.

2. **Hyperparameters are fully specified.** All hyperparameters are documented in Section 2.2: $k = 3$, 10 re-initialisations, tolerance 10^{-4}, 300 max iterations, $\hat{\gamma} = 0.0801$ (RBF, median heuristic), polynomial degree 3, $\gamma_{\text{poly}} = 1.0$, $c_0 = 1.0$, Nyström sizes $m \in \{5, \ldots, 150\}$.

3. **Random seeds are fixed and documented.** Seeds are set as $42 + r$ for run $r = 0, \ldots, 29$, ensuring reproducibility.

4. **Number of runs and statistical tests are reported.** $R = 30$ independent runs per condition; two-sided Welch t-tests with $\alpha = 0.05$; 95% CIs for runtime (Table 5); sensitivity analysis over three DBI margins (Table 4).

5. **Compute resources are documented.** Single CPU core; no GPU. Absolute runtimes are hardware-dependent.

6. **All results tables include uncertainty estimates.** Mean $\pm$ std reported for all quality and cost metrics; 95% CIs for runtime in Table 5.

7. **Data anomalies are reported.** Three anomalies are documented in Section 3.1 with likely explanations and appropriate result caveats.

8. **Limitations are discussed.** Scope limited to $n = 150$ Iris dataset; implementation overhead not separated from algorithmic cost; cosine and large-m anomalies unresolved. See Sections 3.1 and 4.

9. **No human subjects or sensitive data.** The Iris dataset contains no personal information.

10. **Claims are commensurate with evidence.** The original hypothesis of a runtime crossover is explicitly rejected by the data; language is calibrated to reflect absence of evidence vs. evidence of absence throughout.

Confidence Score

Self-assessed Confidence Score: 6.5 / 10.

Rationale: The revised paper addresses all identified critiques—including critical data anomalies, statistical language precision, memory measurement explanation, hypothesis–result alignment, title simplification, and reproducibility documentation—and is internally consistent. The score is not higher because (a) three unresolved data anomalies (Anomaly 1–3) substantially weaken the evidential value of results at the affected sample sizes, (b) the experiment is conducted on a single, very small ($n = 150$) dataset making generalisability claims limited, and (c) figures remain as placeholders. A score above 7 would require re-running experiments to resolve the anomalies and validating at larger dataset scales.